Ryan Strawbridge

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Education

Northeastern University – *Boston, MA* May 2027

B.S. in Mechanical Engineering (BSME) *GPA: 3.58*

M.S. in Mechanical Engineering (MSME) Mechanics & Design Concentration, Expected Dec. 2027, ***Available to work in July 2027***

Coursework: Continuum Mechanics | Heat Transfer | Fluid Mechanics | Solid Mechanics | Thermodynamics | Material Science

Activities: AeroNU Propulsion Member | Running Club President | Rhythm Guitarist in Alt. Rock Band

Achievements: DoD Security Clearance | CSWA Certified | Lean Six Sigma Certification | 800th in the Boston Marathon

Professional Experience

Commonwealth Fusion Systems Devens, MA

Manufacturing / Equipment Engineering Co-op (Term 2) *Jan 2026 – Jun 2026*

- Designed and validated a 2,100-lb hand-portable lifting frame to **ASME BTH-1** and **ASME B30.20** using statics, buckling, threaded-joint, and weld analysis; earned structural engineering and safety approval and **reduced lift time by 30–45 minutes**
- Designed a modular CNC seam-cutting fixture in **Siemens NX** using **ASME Y14.5 GD&T** and dowel-pin datums for repeatable alignment; enabled early qualification testing by de-risking a critical operation
- Designed, fabricated, and deployed a low-cost adjustable fixture using linear rails and carriage assemblies to maintain cable alignment during insulation wrapping, improving process robustness and **reducing labor by 0.5-3 hours per** magnet layer

Manufacturing Engineering Co-op (Term 1) *Jan 2025 – Jun 2025*

- Re-designed a hydraulic piston-driven bending machine in **Siemens NX** using bolt safety analysis and first-principles calculations, cutting operation time and reducing injury risk; **still in use over a year later, saving hundreds of labor hours**
- Developed a predictive equation relating bend offset, bend radius, and arc length through iterative spring-back experiments and data interpolation, **eliminating trial-and-error die design** and enabling future cable-bending dies to be sized analytically on the first iteration
- **Owned a work cell** and wrote detailed work instructions with technicians covering cable assembly, joint placement, orbital cutting, and welding, replacing tribal knowledge with a repeatable reference used on the production floor

Raytheon Missiles & Defense Andover, MA

Operations Engineering Co-op *Jan 2024 – Sep 2024*

- Prototyped a composite fixture using **SolidWorks** and PCB Gerber files to shield heat-sensitive filters from 93 °C while enabling reflow of adjacent components, **increasing first-pass yield** and removing filter damage as a scrap driver on the line
- Ran a nine-configuration solder stencil study varying placement and volume to provide off-gas escape channels, reducing voiding and **removing a rework** step from the process
- **Co-owned a work cell** and developed thermal reflow profiles by refining oven parameters through thermocouple data, operator input, and X-ray inspection to IPC-J-STD-001 Class 3, **reducing rework time across 15 programs**

Projects

Underwater Acoustic Release Mechanism – Capstone Project Boston, MA

Mechanical Design Lead – Release System *Jun 2026 – Present*

- Led mechanical design of a compact underwater release mechanism in **SolidWorks**, owning top-level CAD, **ASME Y14.5 GD&T drawings**, and prototype development for a 35 kg release system rated for 300 m depth
- Engineered the release mechanism and pressure seals using **first-principles analysis** of bending, torsion, O-Ring squeeze, fastener pullout, and seal clamping force; achieved a 2.25 factor of safety on the release arm
- Validated first prototype to 470 psi with zero leakage and **demonstrated successful release** of a 20-lb payload at 60 ft depth after five days submerged in the ocean

AerospaceNU – Liquid Rocket Engine Boston, MA

Propulsion - Feed System Team *Sep 2023 – Present*

- Developing a closed-loop nitrogen pressurant system using a servo-actuated ball valve and downstream pressure feedback for automated pressure regulation
- Supported setup and execution of **three successful static-fire tests**, integrating propulsion, electrical, and instrumentation systems while configuring test hardware and data acquisition equipment

Technical Skills

Software: Siemens NX | SolidWorks | ANSYS Mechanical | Arduino | Teamcenter | SolidWorks PDM | Odoo | Confluence | Jira

Design: Structural Analysis | DFM | GD&T (ASME Y14.5) | FEA

Fabrication: Manual G-Code | CNC Milling | Power Tools | TIG Welding | 3D Printing | Soldering | Water Jetting | Laser Cutting